

## Project Initialization and Planning Phase

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Date          | 15 July 2026                                                   |
| Team ID       | <b>SWUID2026234324</b>                                         |
| Project Title | VideoScriptAI – AI Powered Promotional Video Content Generator |
| Maximum Marks | 5 Marks                                                        |

### Project Proposal (Proposed Solution) report

The proposed solution, **VideoScriptAI**, leverages Generative AI to simplify and accelerate the creation of promotional video content. The application helps marketers, businesses, and content creators generate engaging video hooks, structured scripts, scene suggestions, and compelling call-to-action messages within seconds. By integrating the **Groq API** with the **LLaMA 3.3-70B Versatile** model, the platform delivers high-quality AI-generated content through an intuitive web interface.

| Project Overview  |                                                                                                                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective of VideoScriptAI is to automate promotional video content creation using Generative AI, enabling users to produce professional-quality hooks, scripts, scene ideas, and CTAs quickly and efficiently.                                             |
| Scope             | The project focuses on developing a web-based application using <b>Flask, HTML, CSS, and JavaScript</b> , integrated with the <b>Groq API</b> to generate AI-powered promotional content for products, services, and marketing campaigns.                               |
| Problem Statement |                                                                                                                                                                                                                                                                         |
| Description       | Creating promotional video content manually is time-consuming, requires creativity, and often demands experienced copywriters or marketing professionals. Small businesses and content creators frequently struggle to produce engaging scripts within tight deadlines. |

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|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| Impact                   | By automating promotional content generation, VideoScriptAI significantly reduces content creation time, improves productivity, enhances creativity, and enables businesses to launch marketing campaigns more efficiently.                                                           |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                       |
| Approach                 | Develop a Flask-based web application that securely communicates with the <b>Groq API</b> to generate promotional video content based on user-provided product details, campaign goals, target audience, and video duration.                                                          |
| Key Features             | AI-powered Hook Generator<br>Scene-by-scene Promotional Script Generator<br>Visual Scene Suggestion Generator<br>Call-to-Action (CTA) Generator<br>Generate All feature for complete promotional content creation<br>Secure API key management using .env<br>Responsive web interface |



## Resource Requirements

| Resource Type       | Description                             | Specification/Allocation |
|---------------------|-----------------------------------------|--------------------------|
| <b>Hardware</b>     |                                         |                          |
| Computing Resources | CPU/GPU specifications, number of cores | Standard Laptop/Desktop  |
| Memory              | RAM specifications                      | 8 GB RAM or higher       |

|                         |                                       |                                       |
|-------------------------|---------------------------------------|---------------------------------------|
| Storage                 | Disk space for data, models, and logs | 1 TB SSD                              |
| <b>Software</b>         |                                       |                                       |
| Frameworks              | Python frameworks                     | Flask                                 |
| Libraries               | Additional libraries                  | groq, flask-cors, python-dotenv       |
| Development Environment | IDE                                   | Visual Studio Code                    |
| <b>Data</b>             |                                       |                                       |
| Data                    | Source, size, format                  | Groq API – LLaMA 3.3-70B<br>Versatile |